

XAUUSD AI System Blueprint

Goal

Build a stronger-than-retail XAUUSD trading application that combines:

- NY Opening Range execution
- London momentum and false-break logic
- CPI/NFP/FOMC event mode
- Real risk engine
- Adaptive entries
- Trade journaling
- Daily plan generation
- RAG-backed strategy and context layer

This is not a generic "AI bot." It is a **regime-aware, session-specific gold trading system**.

1) Core Product Positioning

What this system does

The application continuously evaluates XAUUSD across all major trading sessions and decides whether conditions fit one of a small number of high-quality playbooks.

Instead of forcing trades all day, it only trades when:

- the session edge is present
- volatility is acceptable
- spread/slippage are acceptable
- macro risk is understood
- the setup score is above threshold

Primary edges

1. **London Momentum Engine**
 2. **London False-Break Reversal Engine**
 3. **NY Opening Range Engine**
 4. **Event Mode Engine for CPI/NFP/FOMC**
 5. **Adaptive Risk and Execution Engine**
 6. **RAG Layer for context, playbooks, prior trades, and daily planning**
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2) High-Level Architecture

Market Data Feeds

- └ XAUUSD tick / 1m / 5m / 15m / 1h bars
- └ Economic calendar
- └ Spread + slippage metrics
- └ DXY / yields / related macro proxies (optional but recommended)
- └ Broker execution status

Ingestion Layer

- └ Normalize bar data
- └ Build session windows
- └ Detect news windows
- └ Compute indicators/features
- └ Persist raw + derived data

Strategy Layer

- └ Regime Classifier
- └ London Momentum Strategy
- └ London Reversal Strategy
- └ NY Opening Range Strategy
- └ Event Mode Strategy
- └ Adaptive Entry Scoring

Risk Layer

- └ Position sizing
- └ Daily/weekly loss caps
- └ Max session attempts
- └ Drawdown circuit breakers
- └ Spread/slippage guardrails
- └ Prop-firm mode rules

Execution Layer

- └ Broker/MT5 adapter
- └ Order placement
- └ Stop/TP/partial management
- └ Retry / reject handling
- └ Trade state tracking

Intelligence Layer

- └ Daily plan generation
- └ RAG retrieval over playbooks
- └ RAG retrieval over journal history
- └ Strategy explanation engine
- └ End-of-day review generator

Application Layer

- └ API backend
- └ Dashboard
- └ Alerts (Telegram/Discord/Email)
- └ Trade journal UI
- └ Admin / configuration

3) Session Playbooks

A. London Momentum Engine

Objective

Capture genuine range expansion after liquidity enters Europe.

Trading window

- Approx. 02:00–06:00 New York time
- Hard stop window before major red-folder events

Conditions

- Overnight range is defined and reasonably tight
- Volatility is expanding, not collapsing
- Higher-timeframe bias is aligned
- Spread is within threshold

Entry model

- Identify overnight high/low and micro consolidation
- Require momentum confirmation on lower timeframe
- Enter only after breakout is accepted, not first poke
- Reject entries when candle extension is too large versus ATR

Exit model

- Initial stop behind breakout structure or ATR-based
- Partial at 1R
- Trail remainder using micro swing or ATR trail
- Time-based exit if momentum stalls

Best market conditions

- clean directional mornings
- strong London participation

- moderate-to-high volume expansion
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B. London False-Break Reversal Engine

Objective

Exploit gold stop-runs and failed breaks around session transitions.

Conditions

- Sweep of overnight high/low or prior session liquidity
- Immediate rejection and return inside range
- Reversal confirmation on execution timeframe
- No nearby major scheduled news

Entry model

- Wait for liquidity sweep
- Require close back inside range
- Confirm with momentum flip / reclaim of micro structure

Exit model

- Stop beyond the sweep extreme
- TP1 at VWAP/session midpoint
- TP2 at opposite side of range or nearest liquidity pool

Best market conditions

- choppy London open
 - liquidity grabs
 - no strong macro trend continuation
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C. NY Opening Range Engine

Objective

Trade the cleanest continuation or failure from the NY opening range.

Opening range

- Primary OR: 08:20–09:30 New York time

Conditions

- Range boundaries clearly formed
- DXY / macro proxy not strongly contradicting the setup
- No immediate red-folder release unless event mode is active
- Execution quality acceptable

Entry model

- Continuation: break and hold beyond OR high/low with acceptance
- Failure: false break, re-entry back inside range, then reversal toward mid/other side
- Size depends on volatility regime and event proximity

Exit model

- Stop behind OR boundary or micro structure
- Partial at 1R
- Extend runner into prior day high/low or session liquidity target

D. Event Mode Engine (CPI / NFP / FOMC / PPI / Retail Sales / Claims)

Objective

Avoid typical retail-bot destruction around macro events while still allowing a controlled post-release edge.

Rules

- Do not open standard session trades immediately before top-tier releases
- Freeze normal strategies inside a configurable pre/post event buffer
- For event mode, wait for:
 - initial impulse
 - volatility normalization
 - first structured pullback / retest

Event-mode entries

- Trade only in direction of confirmed post-release structure
- Reduce size materially
- Tighter max slippage threshold
- Fewer attempts allowed

Event-mode exits

- Faster partials
- Tighter trailing logic
- Hard flatten if spread/slippage anomalies appear

4) Regime Classifier

The system should not ask "buy or sell?" first. It should ask: **what kind of day is this?**

Target regimes

- Trend day up
- Trend day down
- Range day
- Breakout day
- Reversal/stop-hunt day
- Event day
- Dead/no-trade day

Example features

- Overnight range size
- ATR expansion/contraction
- Session VWAP slope
- Relative distance from prior day high/low
- Candle extension statistics
- Spread quality
- Time of day
- Event proximity
- Optional macro proxies: DXY, US yields

Output

A probability or score for each regime. Only strategies that match the top regime(s) are allowed to activate.

5) Adaptive Entry Scoring

Every setup gets a score instead of a simple yes/no.

Example scoring model

- Higher timeframe bias alignment: 0-2
- Session edge quality: 0-2
- Volatility quality: 0-2
- Liquidity location quality: 0-2
- News risk penalty: -3 to 0
- Spread/slippage penalty: -2 to 0
- Recent execution anomaly penalty: -2 to 0

Trade threshold

Only place trades when total score $\geq$ configured threshold.

Why this is stronger

This reduces overtrading and keeps the system focused on premium conditions.

6) Real Risk Engine

This is a mandatory subsystem, not an optional one.

Core protections

- Fixed fractional risk per trade
- Daily max realized loss
- Weekly max realized loss
- Max open risk exposure
- Max attempts per session
- No martingale
- No averaging down
- No hidden grid behavior
- Auto-disable after abnormal slippage
- Auto-disable after N consecutive setup failures

Prop-firm mode

A special ruleset should include:

- daily drawdown hard cap
- equity protection threshold
- no-trade after daily cap is reached
- reduced risk after first loss of session
- stricter news lockout

Position sizing formula

Position size should be based on:

- account equity
 - risk % per trade
 - stop distance
 - volatility multiplier
 - broker contract specs
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7) Execution Engine

Required features

- Broker adapter (MT5 or broker API)
- Spread filter before order placement
- Max slippage threshold
- Retry policy with cap
- Order reject handling
- Partial profit logic
- Break-even logic
- ATR or structure-based trailing stops
- Forced flat after session cutoff if needed

Why this matters

A mediocre signal with good execution can outperform a strong signal with poor execution.

8) RAG Layer

Use RAG to make the app smarter, more explainable, and more adaptable.

What the RAG system stores

Knowledge collections

1. Strategy Playbooks

2. London momentum rules

3. London reversal rules

4. NY OR rules

5. Event mode rules

6. Risk policies

7. Prop-firm rules

8. Market Context Library

9. Macro event interpretations

10. Gold behavior by session

11. DXY/yield/gold relationships

12. Historical tagged market examples

13. Trade Journal Memory

- 14. all executed trades
- 15. screenshots / chart notes
- 16. outcome summaries
- 17. detected mistakes
- 18. winning pattern notes

19. Operational Docs

- 20. broker settings
- 21. slippage logs
- 22. deployment notes
- 23. troubleshooting

What RAG is used for

- Generate the daily plan
- Explain why a trade is or is not valid
- Retrieve similar historical trades before execution
- Produce end-of-day review
- Highlight repeated mistakes or repeated winning conditions
- Support operator queries like:
 - "Show me similar NY OR failures from the last 60 days"
 - "How did gold behave after CPI beats when DXY also rose?"
 - "What setup type has the highest expectancy in London?"

Example RAG workflow

1. Build market snapshot for current day
 2. Retrieve relevant playbooks
 3. Retrieve similar historical trades
 4. Retrieve nearby macro/event guidance
 5. Ask LLM to generate:
 - 6. daily bias
 - 7. no-trade zones
 - 8. preferred session tactics
 - 9. risk posture
 - 10. explanation
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9) Daily Plan Generation

The app should auto-generate a daily plan before London and before NY.

Inputs

- current XAUUSD structure
- overnight range
- prior day high/low
- session volatility state
- economic calendar
- macro proxies
- recent journal notes
- RAG retrieval results

Output format

- Session bias
- Important levels
- Preferred setup types
- No-trade conditions
- Event mode flag
- Max risk posture for the day
- "What would invalidate the plan?"

Example sections

- Macro snapshot
 - Session map
 - Level map
 - Strategy priority ranking
 - Risk mode
 - Trade checklist
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10) Trade Journaling

Every trade should be automatically journaled.

Journal fields

- timestamp
- session
- strategy type
- regime classification
- setup score
- entry price

- stop price
- target(s)
- realized R multiple
- spread/slippage at entry
- macro context
- screenshot path
- LLM explanation
- operator notes

Auto-review outputs

- top winners by setup type
- top losers by failure reason
- expectancy by session
- expectancy by day of week
- slippage by time of day
- event-mode performance vs normal mode

11) Suggested Technology Stack

Backend

- Python with FastAPI
- PostgreSQL for structured data
- Redis optional for streaming/cache
- Celery or APScheduler for jobs

Market / execution

- MT5 bridge or broker API adapter
- Websocket/stream where available
- Economic calendar feed

AI / RAG

- Embedding model for document and journal chunks
- Vector database for retrieval
- LLM for plan generation, explanation, and review

Frontend

- React / Next.js dashboard
- Session heatmaps
- Daily plan view
- Trade journal explorer

- Risk monitor

Alerts

- Telegram
- Discord
- Email optional

Deployment

- VPS near broker
 - Dockerized services
 - Monitoring + logging
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12) Data Model

Tables

- bars_1m
- bars_5m
- bars_15m
- economic_events
- session_snapshots
- regime_scores
- trade_setups
- orders
- executions
- journal_entries
- daily_plans
- documents
- document_chunks
- embeddings_index
- strategy_backtests

Example journal entry schema

```
{
  "trade_id": "uuid",
  "symbol": "XAUUSD",
  "session": "NY",
  "strategy_type": "ny_opening_range_continuation",
  "regime": "trend_day_up",
  "setup_score": 8,
  "entry_time": "2026-04-21T09:42:00-04:00",
  "entry_price": 3378.2,
```

```

    "stop_price": 3372.9,
    "targets": [3383.5, 3388.0],
    "exit_price": 3385.1,
    "realized_r": 1.3,
    "spread": 0.28,
    "slippage": 0.12,
    "macro_context": ["no_red_folder_now", "dxy_soft"],
    "llm_explanation": "Clean OR break with acceptance above range high;
volatility expansion confirmed.",
    "operator_notes": "Good patience after first fake push.",
    "screenshot_url": "/journal/2026-04-21/trade-uuid.png"
}

```

13) API Surface

Core endpoints

- POST /snapshot/build
- GET /plan/today
- POST /strategy/evaluate
- POST /trade/execute
- POST /trade/flatten
- GET /journal
- GET /journal/{id}
- POST /journal/review
- POST /rag/index
- POST /rag/query
- GET /risk/status
- GET /health

14) Pseudocode: Decision Flow

```

snapshot = build_market_snapshot(symbol="XAUUSD")

if risk_engine.trading_disabled():
    return "NO_TRADE"

if event_engine.in_lockout_window(snapshot):
    if not event_engine.event_mode_enabled(snapshot):
        return "NO_TRADE"

regime_scores = regime_classifier.predict(snapshot)
active_regime = select_regime(regime_scores)

```

```

candidate_setups = []

if active_regime in ["trend_day_up", "trend_day_down", "breakout_day"]:
    candidate_setups += london_momentum.evaluate(snapshot)
    candidate_setups += ny_opening_range.evaluate(snapshot)

if active_regime in ["reversal_day", "range_day"]:
    candidate_setups += london_false_break.evaluate(snapshot)
    candidate_setups += ny_opening_range.evaluate_failure_mode(snapshot)

if active_regime == "event_day":
    candidate_setups += event_mode.evaluate(snapshot)

scored = [score_setup(s, snapshot) for s in candidate_setups]
best = pick_best_valid_setup(scored, threshold=7)

if not best:
    return "NO_TRADE"

order = risk_engine.size_and_validate(best, snapshot)
execution = execution_engine.place(order)
journal_engine.log_execution(best, execution, snapshot)
return execution

```

15) RAG Prompt Design

Daily plan prompt skeleton

You are the XAUUSD desk strategist.
Use the retrieved playbooks, current market snapshot, economic calendar, and similar historical journal entries.

Tasks:

1. Classify today's gold environment.
2. Rank best setup types for London and NY.
3. Identify key levels.
4. Define no-trade conditions.
5. Set risk posture for the day.
6. Produce a concise execution checklist.

Rules:

- Prefer capital preservation when signals conflict.
- Respect event lockout windows.

- Do not recommend trades without clear session edge.
- Keep output structured for the dashboard.

Similar-trade retrieval examples

- Retrieve top 10 historical London momentum trades with current ATR range and no major news
 - Retrieve failed NY OR continuation trades when spread > threshold
 - Retrieve event-mode trades similar to current CPI/NFP setup
-

16) Build Order

Phase 1: Foundation

- market data ingestion
- session detection
- risk engine
- manual dashboard
- journal database

Phase 2: Core strategies

- London momentum
- London reversal
- NY OR
- event lockout

Phase 3: RAG intelligence

- playbook indexing
- journal retrieval
- daily plan generation
- trade explanation layer

Phase 4: Hardening

- backtests
- walk-forward validation
- Monte Carlo stress tests
- execution anomaly detection
- broker comparison

Phase 5: Production automation

- VPS deployment
- alerts
- session schedulers
- auto-review and reporting

17) Validation Standards

Before trusting live capital, require:

- out-of-sample tests
- walk-forward tests
- Monte Carlo equity stress
- expectancy by setup type
- expectancy by session
- live/demo forward test
- slippage/spread audit
- broker sensitivity analysis

Required metrics

- win rate
- profit factor
- expectancy per trade
- max drawdown
- average MAE/MFE
- R distribution
- time-in-trade
- session breakdown
- event-mode stats

18) Why This Can Outperform Generic Retail Bots

Because it is designed to:

- trade fewer but better spots
- understand session context
- respect macro events
- adapt risk to conditions
- learn from its own journal memory
- explain itself through RAG
- preserve capital when the market is hostile

That is a stronger design than a generic all-day indicator bot.

19) Minimum Viable Product Definition

MVP includes

- XAUUSD only
- London momentum
- NY OR continuation/failure
- event lockout
- risk engine
- dashboard
- journal
- daily plan generation with RAG

MVP excludes initially

- too many instruments
 - overcomplicated ML
 - portfolio allocation
 - fully autonomous retraining
-

20) Final Recommendation

The best version of this product is a **hybrid intelligence trading system**, not a black-box robot.

Use deterministic strategy engines for execution. Use RAG + LLM for:

- daily planning
- contextual explanation
- historical memory retrieval
- post-trade review
- operator assistance

This gives you a system that is stronger, more transparent, and safer than typical retail bots.